

SMARTEDIT: A HUMAN-IN-THE-LOOP AI FRAMEWORK FOR MULTI-FOOTAGE VIDEO EDITING

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Mini Project

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- ▶ Video editing is a time-consuming process because editors have to watch all the raw videos, remove unwanted parts, and arrange clips manually.
- ▶ When there are many video clips, these tasks become repetitive and increase the editing time.
- ▶ AI can help with video editing, but it cannot fully understand the editor's creative ideas or editing style.

Introduction

- ▶ Creating the first version of a video takes a lot of time because editors have to find the best clips, remove shaky videos, trim silent parts, and arrange the clips.
- ▶ AI can edit videos quickly, but it cannot fully understand the editor's ideas, storytelling style, or personal preferences.
- ▶ Human-in-the-Loop (HITL) AI combines the speed of automation with human decision-making, letting the editor review, accept, reject, or modify AI suggestions before finalizing the output.

Objectives

Develop a video editing system that automates repetitive editing tasks using NLP and AI analysis while keeping editors in complete creative control.

- ▶ Interpret natural language editing prompts to understand user editing requirements.
- ▶ Analyze video and audio content to identify key scenes and editing opportunities.
- ▶ Generate an AI-assisted rough-cut timeline with recommended edits.
- ▶ Allow editors to preview, accept, reject, or manually refine AI-generated suggestions.
- ▶ Improve editing efficiency by reducing manual effort while preserving creative control.

Tilekbay, T. et al., “ExpressEdit: A Multimodal System for Video Editing with Natural Language and Sketching,” *Proc. ACM CHI Conference on Human Factors in Computing Systems*, 2024.

- ▶ Enables users to edit videos through natural language prompts combined with sketch-based interactions, improving accessibility and user interaction.
- ▶ **Observation:** Supports editing commands but provides no AI-assisted rough-cut generation, computer vision analysis, or Human-in-the-Loop review – the gap SmartEdit targets.

Chen, Z. et al., “EditBoard: A Comprehensive Evaluation Benchmark for Text-based Video Editing,” *arXiv preprint arXiv:2402.09257*, 2025.

- ▶ Introduces a comprehensive benchmark for evaluating text-based AI video editing models and comparing their performance.
- ▶ **Observation:** Useful for evaluation, but offers no interactive editing system – confirming the need for a working, user-facing HITL application rather than a benchmark alone.

Wang, K. et al., “LAVE: An LLM-powered Agent for Video Editing,” *arXiv preprint arXiv:2402.19131*, 2024.

- ▶ Uses Large Language Models to understand user instructions and automate video editing tasks.
- ▶ **Observation:** Automates editing well but gives limited user control during editing and lacks Human-in-the-Loop refinement – motivating SmartEdit’s review-before-export workflow.

Shekar, K. et al., “AI-Based Intelligent Video Editing System,” *Proc. Int. Conf. Advancements in Smart, Secure and Intelligent Computing (ASSIC)*, 2025.

- ▶ Proposes an AI-assisted framework for automatic scene selection and video editing.
- ▶ **Observation:** Does not integrate natural language prompt interpretation, interactive timeline editing, or user-guided refinement – all core to SmartEdit’s design.

Gautham, B. and Somasundaram, K., “Social Media Content Generation: An AI-Based System for Automated Viral Short-Form Video Creation,” *Proc. IEEE Int. Conf. on Information Technology, Electronics and Intelligent Communication Systems (ICITEICS)*, 2025.

- ▶ Presents a fully automated LLM-based pipeline that downloads long-form YouTube video, transcribes it, and uses a multimodal LLM to detect “viral” moments, which are then cropped to 9:16 and captioned for Shorts/Reels/TikTok.
- ▶ **Observation:** Strong end-to-end automation for clip selection and captioning, but offers no natural-language rough-cut authoring, no shaky/blur-based footage triage, and no Human-in-the-Loop review step before export – gaps that motivate SmartEdit’s editor-in-the-loop workflow.

Reference	Approach	Brief Observation
Tilekbay et al. (2024)	NL + sketch video editing	No AI rough-cut generation or HITL review.
Chen et al. (2025)	Text-based editing benchmark	Evaluation only, not an interactive system.
Wang et al. (2024)	LLM video editing agent	Automates editing with limited user control.
Shekar et al. (2025)	AI scene selection & editing	No NL prompts or interactive refinement.
Gautham & Somasundaram (2025)	Auto-finds, crops viral clips	Checks footage quality (shaky/blur), takes user prompts, and needs human approval – this system is fully hands-off.

- ▶ **Adobe Premiere Pro:** A powerful video editing software with AI features, but editors still need to manually choose clips and create the first rough cut.
- ▶ **DaVinci Resolve:** Uses AI for tasks like object detection and face recognition, but it cannot edit videos based on simple text instructions.
- ▶ **Runway ML:** Uses AI to edit videos automatically, but users have limited control over the editing decisions.
- ▶ **Common Limitations:**
 - ▶ Video editing still takes a lot of time.
 - ▶ AI cannot fully understand the editor's creative ideas.
 - ▶ Limited user control over AI-generated edits.
 - ▶ AI features are not combined into one complete editing system.

Proposed System

- ▶ **SmartEdit:** an AI-assisted Human-in-the-Loop video editing application built with Qt (C++), Python, OpenCV, Librosa, and FFmpeg.
- ▶ **Prompt Interpretation Module:** A Small Language Model (SLM) converts natural language editing instructions into structured editing commands.
- ▶ **Video Analysis Module:** OpenCV detects shaky footage, blurred scenes, and duplicate/low-quality clips.
- ▶ **Audio Analysis Module:** Librosa detects silent regions and low-energy segments for automatic trimming.
- ▶ **AI-Assisted Rough Cut Generation:** Combines prompt, video, and audio analysis into a preliminary timeline matching the requested style and duration.
- ▶ **Human-in-the-Loop Review Module:** Users preview, accept, reject, and rearrange AI suggestions before anything is finalized.
- ▶ **Video Rendering and Export Module:** FFmpeg encodes and exports the finalized timeline at high quality.

Hardware Requirements

- ▶ **Processor:** Intel Core i5 (11th Gen or higher) / AMD Ryzen 5 or higher
- ▶ **Memory:** Minimum 8 GB RAM (16 GB recommended)
- ▶ **Storage:** Minimum 100 GB SSD free space
- ▶ **Graphics:** NVIDIA GeForce GTX 1650 or higher
- ▶ **Display:** Full HD (1920 × 1080) monitor

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Requirements Specification

Software Requirements

- ▶ **Operating System:** Windows 10/11 (64-bit); adaptable to Linux with Qt support
- ▶ **Languages:** C++ , Python 3.11 or later (AI modules)
- ▶ **GUI Framework:** Qt 6 – interface, timeline editor, media playback
- ▶ **Computer Vision:** OpenCV – shaky footage/frame-quality detection
- ▶ **Audio Processing:** Librosa – silence detection, feature extraction
- ▶ **AI Model:** Small Language Model (SLM)
- ▶ **Multimedia Engine:** FFmpeg – decoding, trimming, encoding, export
- ▶ **Dev Tools:** Qt Creator, Visual Studio Code, Git, Python virtual environments

Requirements Specification

Functional Requirements

- ▶ **Video Import & Project Creation:** Import multiple video files; validate format; extract metadata (duration, resolution, frame rate, audio).
- ▶ **Prompt-Based Editing:** Users specify style, target duration, clip arrangement, and preferences (e.g. remove silence/shaky footage) in natural language; SLM converts these into structured instructions.
- ▶ **Video Analysis:** Detect shaky footage, blurred/low-quality frames, and duplicate clips via OpenCV.
- ▶ **Audio Analysis:** Detect silence, low-energy segments, and speech activity via Librosa.
- ▶ **AI-Assisted Rough Cut Generation:** Arrange clips per prompt, remove unnecessary/silent footage, and match target duration.

- ▶ **Human-in-the-Loop Review:** Preview, accept, reject, or regenerate the rough cut – no edit is applied without approval.
- ▶ **Manual Timeline Editing:** Trim, rearrange, insert, or remove clips after the rough cut is approved.
- ▶ **Rendering & Export:** FFmpeg-based export with progress tracking and quality/sync preservation.

Non-Functional Requirements

- ▶ **Performance:** Responsive playback and low-latency AI analysis even for high-resolution video, evaluated using the metrics on the following slide.
- ▶ **Usability:** Clean, intuitive interface suited to both beginners and professional editors.
- ▶ **Accuracy & Reliability:** Consistent detection and suggestions; no permanent edit without user approval.
- ▶ **Security & Data Privacy:** Project files accessible only to authorized users; no unconfirmed modification/deletion.
- ▶ **Maintainability & Extensibility:** Modular architecture separating UI, AI modules, and rendering engine.
- ▶ **Scalability:** Supports multiple video files of varying resolution/duration within one project.

Performance Metrics

Clip Selection Precision $\frac{\text{Relevant selected}}{\text{Total selected}} \times 100$ How many AI-selected clips are actually useful

Clip Selection Recall $\frac{\text{Relevant selected}}{\text{Total relevant}} \times 100$ How many useful clips the AI successfully selects

F1-Score $2 \times \frac{P \times R}{P + R}$ Combined measure of precision and recall

Temporal IoU $\frac{\text{Intersection}}{\text{Union}}$ Accuracy of selected time-range boundaries

Clip Order Accuracy $\frac{\text{Correctly ordered}}{\text{Total clips}} \times 100$ Accuracy of clip arrangement vs. reference timeline

Duration Error $\frac{|D_{AI} - D_{Ref}|}{D_{Ref}} \times 100$ Deviation of AI rough-cut duration from reference

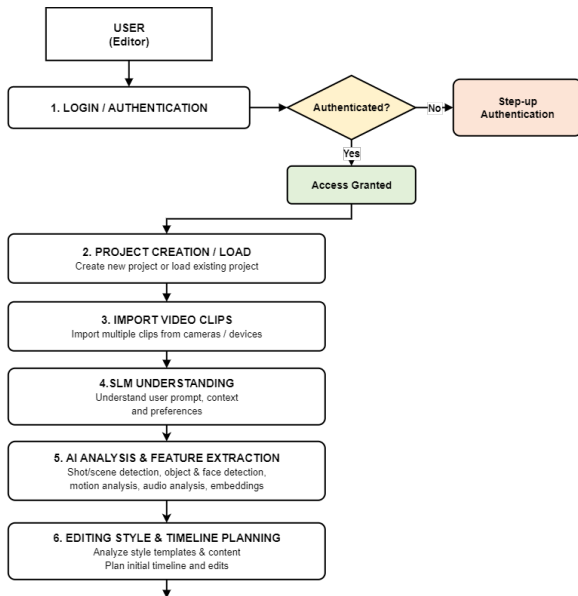
Edit Decision Accuracy $\frac{\text{Correct decisions}}{\text{Total decisions}} \times 100$ Accuracy of trim/remove/keep/arrange decisions

Human Acceptance Rate $\frac{\text{Accepted suggestions}}{\text{Total suggestions}} \times 100$ How often editors accept AI suggestions

Input and Output

- ▶ **Input:** One or more raw video files imported from local storage, plus a natural language prompt describing editing style and target duration.
- ▶ **Input Format:** Standard multimedia video files supported by OpenCV/FFmpeg.
- ▶ **Output:** An AI-generated, reviewable rough-cut timeline followed by a final rendered video reflecting user-approved and manually refined edits.
- ▶ **Output Format:** Exported video file in a supported format, produced via FFmpeg with preserved video/audio quality and synchronization.

System Architecture



System Architecture

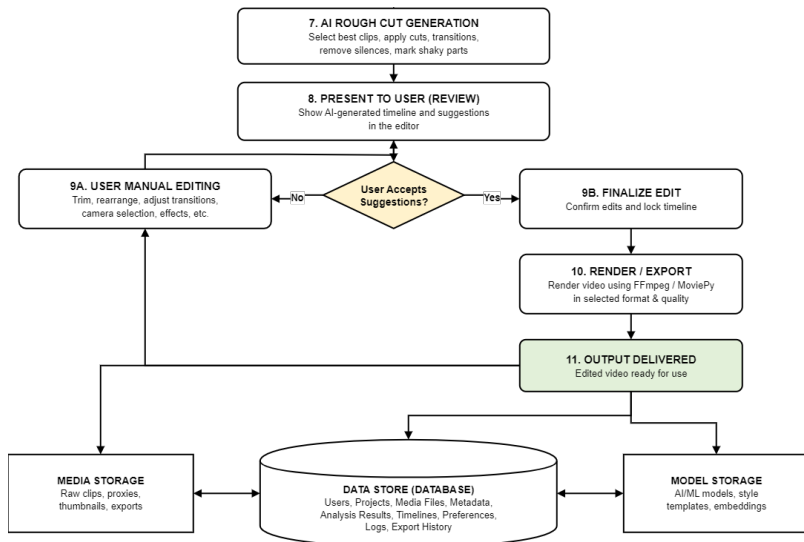


Figure 1: SmartEdit AI System Architecture

Data Flow Diagrams

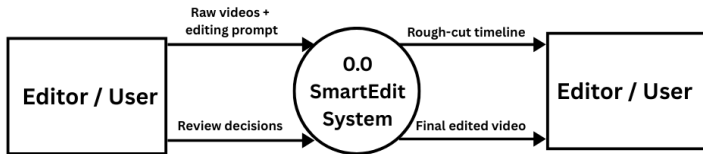


Figure 2: Level 0 DFD

Data Flow Diagrams

Data Flow Diagram – Level 1 (SmartEdit)

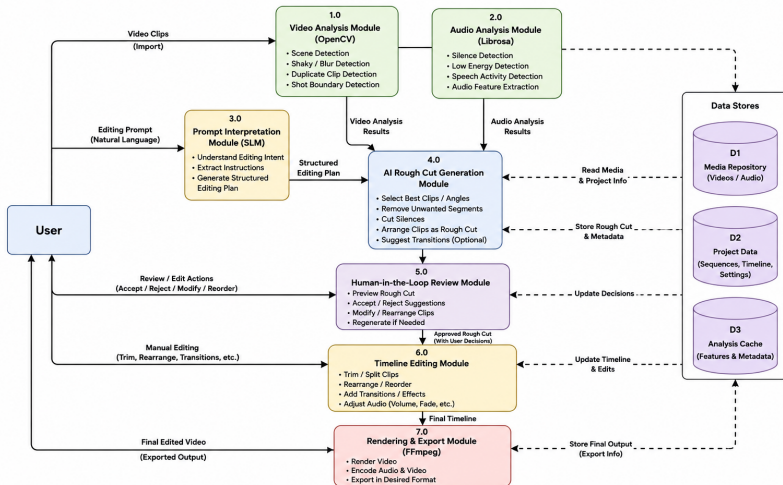


Figure 3: Level 1 DFD

- ▶ **User Interface Module:** Provides a desktop interface built with Qt for importing videos, entering natural language editing prompts, previewing clips, and managing the editing timeline.
- ▶ **Project Management Module:** Creates and manages editing projects, stores project metadata, organizes imported media, and maintains editing sessions.
- ▶ **Prompt Interpretation Module:** Uses a Small Language Model (SLM) to convert natural language editing instructions into structured editing commands.
- ▶ **Video Analysis Module:** Analyzes imported videos using OpenCV through:
 - ▶ Scene Detection — Identifies scene boundaries.
 - ▶ Quality Analysis — Detects shaky footage, blurred frames, and duplicate clips.
 - ▶ Metadata Extraction — Extracts video properties.

System Modules

- ▶ **Audio Analysis Module:** Uses Librosa to detect silence, low-energy segments, and speech activity for automatic trimming.
- ▶ **AI Rough Cut Generation Module:** Combines prompt interpretation, video analysis, and audio analysis to generate an editable rough-cut timeline.
- ▶ **Human-in-the-Loop Review Module:** Allows the editor to preview, accept, reject, modify, rearrange, or regenerate AI-generated suggestions before finalizing the edit.
- ▶ **Timeline Editing Module:** Provides manual tools for trimming clips, rearranging the timeline, adding transitions, and refining the rough cut.
- ▶ **Rendering and Export Module:** Uses FFmpeg to render the approved timeline and export the final edited video while preserving video and audio quality.

- ▶ SmartEdit combines natural language prompt interpretation, computer vision, and audio signal processing within a single Human-in-the-Loop desktop application.
- ▶ Unlike existing fully automated or purely evaluative tools, it focuses on generating a reviewable rough cut that the editor can accept, reject, or refine before export.
- ▶ Aims to significantly reduce rough-cut creation time while keeping complete creative control with the human editor.
- ▶ Built on mature open-source technologies, making it practical and affordable for content creators and media professionals.

References

- [1] T. Tilekbay, A. Kim, D. E. Park and J. A. Landay, “ExpressEdit: A Multimodal System for Video Editing with Natural Language and Sketching,” in *Proc. ACM CHI Conference on Human Factors in Computing Systems (CHI)*, 2024.
- [2] Z. Chen, K. Liang, H. Wang et al., “EditBoard: A Comprehensive Evaluation Benchmark for Text-based Video Editing,” *arXiv preprint arXiv:2402.09257*, 2025.
- [3] K. Wang, Z. Wang, Z. Yuan et al., “LAVE: An LLM-powered Agent for Video Editing,” *arXiv preprint arXiv:2402.19131*, 2024.
- [4] K. Shekar, K. Kasi Viswanadh, V. Srujan Varma, K. Veda Sri, S. Aditya and S. Sahoo, “AI-Based Intelligent Video Editing System,” in *2025 Int. Conf. Advancements in Smart, Secure and Intelligent Computing (ASSIC)*, 2025.

- [5] G. B. Gautham and K. Somasundaram, "Social Media Content Generation: An AI-Based System for Automated Viral Short-Form Video Creation," in 2025 IEEE 2nd International Conference on Information Technology, Electronics and Intelligent Communication Systems (ICITEICS), 2025.

Thank You